

Problem Set 1

Wen Sheng (ws2840)

2026-10-05

Problem 1

(i)

By Definition,

$$\begin{aligned} E[Y|X = x] &= \sum_y y \times p(y|X = x) \\ &= \sum_y y \times \frac{P(X = x, Y = y)}{P(X = x)} \end{aligned}$$

The joint probability distribution $P(X = x, Y = y)$ is:

$X \backslash Y$	2	3	4	5	6	7	8	9	10	11	12	$P(X = x)$
1	$\frac{1}{36}$	0	0	0	0	0	0	0	0	0	0	$\frac{1}{36}$
2	0	$\frac{2}{36}$	$\frac{1}{36}$	0	0	0	0	0	0	0	0	$\frac{3}{36}$
3	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	0	0	0	0	$\frac{5}{36}$
4	0	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	0	0	$\frac{7}{36}$
5	0	0	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	$\frac{9}{36}$
6	0	0	0	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{11}{36}$
$P(Y = y)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	1

So,

$$E[Y|X = 1] = 2 * \frac{\frac{1}{36}}{\frac{1}{36}} = 2$$

$$E[Y|X = 2] = 3 * \frac{\frac{2}{36}}{\frac{2}{36}} + 4 * \frac{\frac{1}{36}}{\frac{2}{36}} = \frac{10}{3}$$

$$E[Y|X = 3] = 4 * \frac{\frac{2}{36}}{\frac{5}{36}} + 5 * \frac{\frac{2}{36}}{\frac{5}{36}} + 6 * \frac{\frac{1}{36}}{\frac{5}{36}} = \frac{24}{5}$$

$$E[Y|X = 4] = 5 * \frac{\frac{2}{36}}{\frac{7}{36}} + 6 * \frac{\frac{2}{36}}{\frac{7}{36}} + 7 * \frac{\frac{2}{36}}{\frac{7}{36}} + 8 * \frac{\frac{1}{36}}{\frac{7}{36}} = \frac{44}{7}$$

$$E[Y|X = 5] = 6 * \frac{\frac{2}{36}}{\frac{9}{36}} + 7 * \frac{\frac{2}{36}}{\frac{9}{36}} + 8 * \frac{\frac{2}{36}}{\frac{9}{36}} + 9 * \frac{\frac{2}{36}}{\frac{9}{36}} + 10 * \frac{\frac{1}{36}}{\frac{9}{36}} = \frac{70}{9}$$

$$E[Y|X = 6] = 7 * \frac{\frac{2}{36}}{\frac{11}{36}} + 8 * \frac{\frac{2}{36}}{\frac{11}{36}} + 9 * \frac{\frac{2}{36}}{\frac{11}{36}} + 10 * \frac{\frac{2}{36}}{\frac{11}{36}} + 11 * \frac{\frac{2}{36}}{\frac{11}{36}} + 12 * \frac{\frac{1}{36}}{\frac{11}{36}} = \frac{102}{11}$$

(ii)

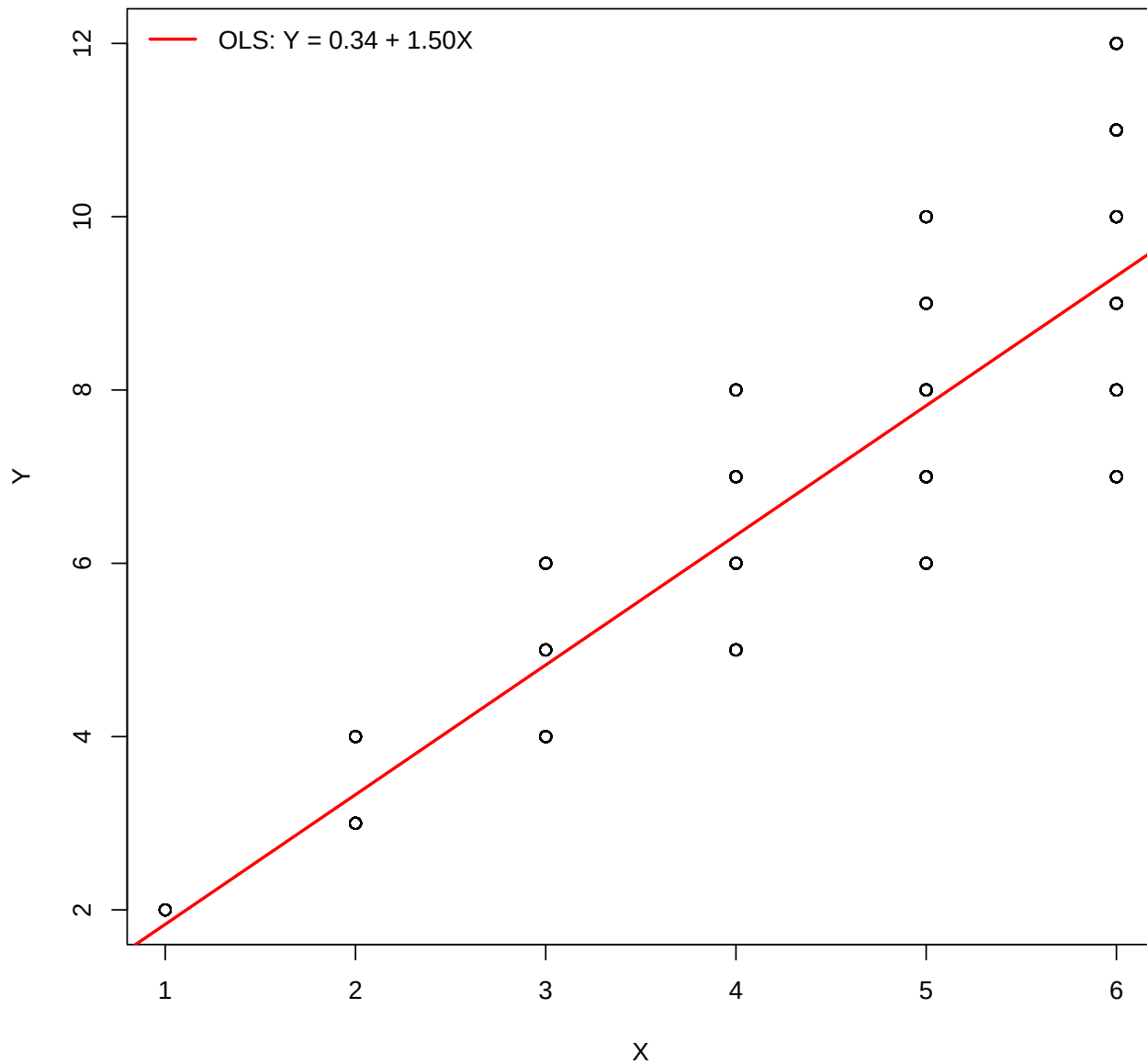
Simulate $n = 1000$ draws of (X, Y) . Plot the draws:

```
set.seed(17)
n <- 1000
D1 <- sample(1:6, n, replace = TRUE)
D2 <- sample(1:6, n, replace = TRUE)

X <- pmax(D1, D2)
Y <- D1 + D2
ols <- lm(Y ~ X)

plot(X, Y, main = "Sum and Max of Two Dice Rolls")
abline(ols, col = "red", lwd = 2)
legend("topleft",
      sprintf("OLS: Y = %.2f + %.2fX", coef(ols)[1], coef(ols)[2]),
      col = "red", lwd = 2, bty = "n")
```

Sum and Max of Two Dice Rolls



(iii)

After simulation $n = 1000$ observations and fitting the OLS regression,

$$\hat{\beta}_0 = 0.34, \quad \hat{\beta}_1 = 1.50$$

so

$$E[Y|\widehat{X} = x] = \hat{\beta}_0 + \hat{\beta}_1 x = 0.34 + 1.50x$$

(iv)

Using the estimated line, we get:

x	True	OLS
1	2.000	1.84
2	3.333	3.34
3	4.800	4.84
4	6.286	6.34
5	7.778	7.84
6	9.273	9.34

The linear predictor is **biased** for $E[Y | X = x]$ because the true conditional expectation is nonlinear in x .

(v)

Simulate and estimated $B(x)$:

```
M <- 2000
N <- 1000

x_vals <- 1:6
y_vals <- 2:12
join_prob <- matrix(0, nrow = 6, ncol = 11, dimnames = list(x_vals, y_vals))
join_prob["1", "2"] <- 1
join_prob["2", c("3", "4")] <- c(2, 1)
join_prob["3", c("4", "5", "6")] <- c(2, 2, 1)
join_prob["4", c("5", "6", "7", "8")] <- c(2, 2, 2, 1)
join_prob["5", c("6", "7", "8", "9", "10")] <- c(2, 2, 2, 2, 1)
join_prob["6", c("7", "8", "9", "10", "11", "12")] <- c(2, 2, 2, 2, 2, 1)
join_prob <- join_prob / 36

# calculate true E[Y|X=x]
P_X <- rowSums(join_prob)
true_EY_given_X <- as.vector(join_prob %*% y_vals) / P_X

# simulation: M=2000 times N=1000 draws
Yhat_mat <- matrix(NA, nrow = M, ncol = length(x_vals))
for (r in 1:M) {
  D1_r <- sample(1:6, N, replace = TRUE)
  D2_r <- sample(1:6, N, replace = TRUE)
  X_r <- pmax(D1_r, D2_r)
  Y_r <- D1_r + D2_r
  fit_r <- lm(Y_r ~ X_r)
  Yhat_mat[r, ] <- coef(fit_r)[1] + coef(fit_r)[2] * x_vals
}
EYhat <- colMeans(Yhat_mat)
```

```

Bias_x <- EYhat - true_EY_given_X
names(Bias_x) <- x_vals
bias_table <- data.frame(
  x = x_vals,
  true_EY = true_EY_given_X,
  E_Yhat = EYhat,
  Bias = Bias_x
)
bias_table

```

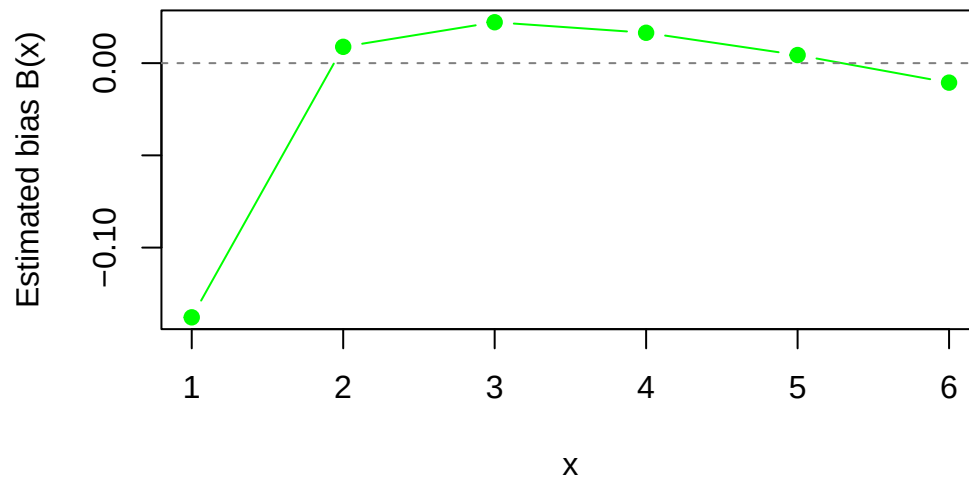
	x	true_EY	E_Yhat	Bias
1	1	2.000000	1.862189	-0.137811003
2	2	3.333333	3.342184	0.008850194
3	3	4.800000	4.822178	0.022178058
4	4	6.285714	6.302173	0.016458303
5	5	7.777778	7.782167	0.004389341
6	6	9.272727	9.262162	-0.010565623

```

plot(x_vals, Bias_x, type = "b", pch = 19, col = "green",
     xlab = "x", ylab = "Estimated bias B(x)",
     main = "Estimated bias of the linear predictor Yhat(x)",
     abline(h = 0, lty = 2, col = "gray50"))

```

Estimated bias of the linear predictor Yhat(x)



Problem 2

Simplify the conditional density of $Y|X = x$:

$$f(y|X = x) = \frac{f(x, y)}{f_X(x)}$$

Let

$$z_x = \frac{x - \mu_x}{\sigma_x}, z_y = \frac{y - \mu_y}{\sigma_y}$$

Then

$$\begin{aligned} f(x, y) &= \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} [z_x^2 - 2\rho z_x z_y + z_y^2] \right\} \\ &= \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} [z_y^2 - 2\rho z_x z_y + \rho^2 z_x^2 - \rho^2 z_x^2 + z_x^2] \right\} \\ &= \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} (z_y - \rho z_x)^2 - \frac{1}{2} z_x^2 \right\} \\ &= \frac{1}{\sqrt{2\pi\sigma_y^2(1-\rho^2)}} \exp \left\{ -\frac{1}{2} \left(\frac{z_y - \rho z_x}{\sqrt{1-\rho^2}} \right)^2 \right\} \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp \left\{ -\frac{1}{2} z_x^2 \right\} \\ &= h(x, y)g(x) \end{aligned}$$

Which $h(x, y)$ is also a valid normal density function:

$$\begin{aligned} h(x, y) &= \frac{1}{\sqrt{2\pi\sigma_y^2(1-\rho^2)}} \exp \left\{ -\frac{1}{2} \left(\frac{z_y - \rho z_x}{\sqrt{1-\rho^2}} \right)^2 \right\} \\ &= \frac{1}{\sqrt{2\pi\sigma_y^2(1-\rho^2)}} \exp \left\{ -\frac{1}{2} \left(\frac{\frac{y-\mu_y}{\sigma_y} - \rho \frac{x-\mu_x}{\sigma_x}}{\sqrt{1-\rho^2}} \right)^2 \right\} \\ &= \frac{1}{\sqrt{2\pi\sigma_y^2(1-\rho^2)}} \exp \left\{ -\frac{1}{2} \left(\frac{y - \mu_y - \rho\sigma_y \frac{x-\mu_x}{\sigma_x}}{\sigma_y\sqrt{1-\rho^2}} \right)^2 \right\} \end{aligned}$$

The marginal distribution of X:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp \left\{ -\frac{1}{2} z_x^2 \right\} h(x, y) dy \\ &= \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp \left\{ -\frac{1}{2} z_x^2 \right\} \int_{-\infty}^{\infty} h(x, y) dy \\ &= \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp \left\{ -\frac{1}{2} z_x^2 \right\} \end{aligned}$$

Thefore, $f_X(x) = g(x)$

Now we have,

$$\begin{aligned} f(y|X=x) &= \frac{f(x,y)}{f_X(x)} \\ &= \frac{h(x,y)g(x)}{g(x)} \\ &= h(x,y) \\ &= \frac{1}{\sqrt{2\pi\sigma_y^2(1-\rho^2)}} \exp \left\{ -\frac{1}{2} \left(\frac{y - \mu_y - \rho\sigma_y \frac{x-\mu_x}{\sigma_x}}{\sigma_y \sqrt{1-\rho^2}} \right)^2 \right\} \end{aligned}$$

Hence,

$$Y|X \sim N(\mu_y + \rho \frac{\sigma_y}{\sigma_x}(x - \mu_x), \sigma_y^2(1 - \rho^2))$$

Problem 3

(i)

Fit OLS R Code:

```
model <- lm(y ~ x)
model
```

Call:

```
lm(formula = y ~ x)
```

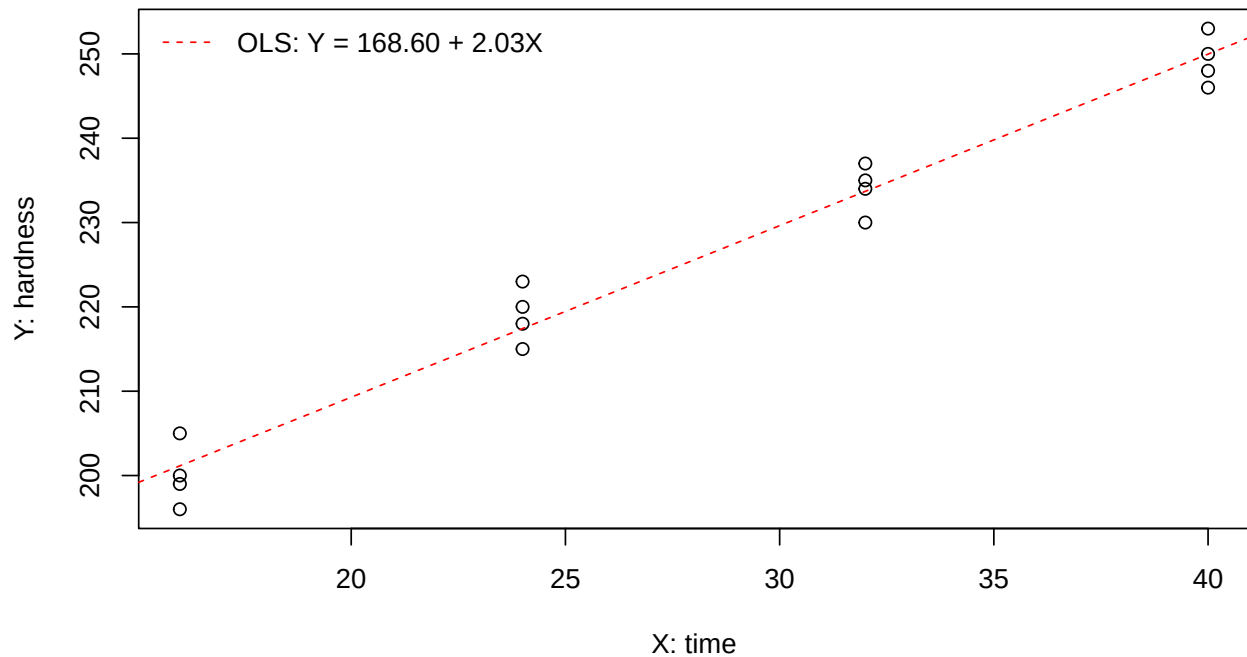
Coefficients:

(Intercept)	x
168.600	2.034

(ii)

We plot hardness against time for the 16 observations and overlay the fitted OLS regression line.

```
plot(x, y, xlab = "X: time", ylab = "Y: hardness")
abline(model, col = "red", lty = 2)
legend("topleft",
       sprintf("OLS: Y = %.2f + %.2fX", coef(model)[1], coef(model)[2]),
       col = "red", lty = 2, bty = "n")
```



(iii)

Compute MSE,

$$MSE = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = 10.45893$$

```
y_hat <- fitted(model)
SSE <- sum((y - y_hat)^2)
MSE <- SSE / (length(y) - 2)
MSE
```

```
[1] 10.45893
```

(iv)

The sample correlation coefficient is:

$$corr = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} = 0.9865$$

```
cor(x, y)
```

```
[1] 0.9864599
```

The coefficient of determination is:

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 0.9731$$

```
SST <- sum((y - mean(y))^2)
R2 <- 1 - SSE / SST
R2
```

[1] 0.9731031

Problem 4

(i)

Proof. The least-squares estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are obtained by minimizing the sum of squared residuals:

$$Q(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

over $b_0, b_1 \in \mathbb{R}$. Because Q is differentiable on $\mathbb{R}^2$, the minimum occurs where the partial derivatives with respect to b_0 and b_1 are zero.

$$\begin{aligned} 0 &= \left. \frac{\partial Q}{\partial b_0} \right|_{b_0=\hat{\beta}_0, b_1=\hat{\beta}_1} \\ &= -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \\ &= -2 \sum_{i=1}^n e_i \end{aligned}$$

Hence, the residuals sum to zero:

$$\sum_{i=1}^n e_i = 0$$

Similarly, differentiating with respect to the slope gives

$$\frac{\partial Q}{\partial b_1} = -2 \sum_{i=1}^n x_i (y_i - b_0 - b_1 x_i) = 0$$

Solving these equations gives the least-squares estimates:

$$\begin{aligned} 0 &= \left. \frac{\partial Q}{\partial b_1} \right|_{b_0=\hat{\beta}_0, b_1=\hat{\beta}_1} \\ &= -2 \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \\ &= -2 \sum_{i=1}^n x_i e_i \end{aligned}$$

Therefore,

$$\sum_{i=1}^n x_i e_i = 0$$

which proves part(i).

(ii)

Proof. Use the fitted value for the i -th observation, we have

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

and the two normal equations in Part (i). Then

$$\begin{aligned} \sum_{i=1}^n \hat{y}_i e_i &= \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i) e_i \\ &= \hat{\beta}_0 \sum_{i=1}^n e_i + \hat{\beta}_1 \sum_{i=1}^n x_i e_i \\ &= 0 \end{aligned}$$

Thus, the residual vector is orthogonal to the fitted-value vector, which proves part (ii).

Problem 5

(i)

Proof.

$$\begin{aligned} \sum_{j=1}^n h_{ij}^2 &= \sum_{j=1}^n \left(\frac{1}{n} + \frac{(x_i - \bar{x})(x_j - \bar{x})}{S_{xx}} \right)^2 \\ &= \sum_{j=1}^n \left(\frac{1}{n^2} + \frac{2(x_i - \bar{x})(x_j - \bar{x})}{nS_{xx}} + \frac{(x_i - \bar{x})^2(x_j - \bar{x})^2}{S_{xx}^2} \right) \\ &= \sum_{j=1}^n \frac{1}{n^2} + \frac{2(x_i - \bar{x})}{nS_{xx}} \sum_{j=1}^n (x_j - \bar{x}) + \frac{(x_i - \bar{x})^2}{S_{xx}^2} \sum_{j=1}^n (x_j - \bar{x})^2 \\ &= \frac{1}{n} + 0 + \frac{(x_i - \bar{x})^2}{S_{xx}^2} S_{xx} \\ &= \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}} \\ &= h_{ii} \end{aligned}$$

Hence, the sum of the squares of i -th row of the hat matrix: $\sum_{j=1}^n h_{ij}^2$ equals the diagonal element h_{ii} .

(ii)

Proof.

$$\begin{aligned}\sum_{j=1}^n h_{ij}x_j &= \sum_{j=1}^n \left(\frac{1}{n} + \frac{(x_i - \bar{x})(x_j - \bar{x})}{S_{xx}} \right) x_j \\ &= \frac{1}{n} \sum_{j=1}^n x_j + \frac{(x_i - \bar{x})}{S_{xx}} \sum_{j=1}^n (x_j - \bar{x})x_j \\ &= \bar{x} + \frac{(x_i - \bar{x})}{S_{xx}} S_{xx} \\ &= \bar{x} + (x_i - \bar{x}) \\ &= x_i\end{aligned}$$

Hence, the dot product of the i -th row of the hat matrix with the vector of x values: $\sum_{j=1}^n h_{ij}x_j$, equals the i -th x value itself.

Problem 6

(i)

Under model (1),

$$E[Y_i] = E[\beta x_i + \epsilon_i] = \beta x_i + E[\epsilon_i] = \beta x_i$$

Then,

$$\begin{aligned}E[\hat{\beta}] &= E \left[\frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n x_i^2} \right] \\ &= \frac{\sum_{i=1}^n x_i E[Y_i]}{\sum_{i=1}^n x_i^2} \\ &= \frac{\sum_{i=1}^n x_i (\beta x_i)}{\sum_{i=1}^n x_i^2} \\ &= \frac{\beta \sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2} \\ &= \beta\end{aligned}$$

Hence, $\hat{\beta}$ is unbiased for β .

(ii)

Solution. Note, $\hat{\beta}$ can be written as a linear combination of the Y_i 's:

$$\begin{aligned}\hat{\beta} &= \frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n (x_i)^2} \\ &= \sum_{i=1}^n c_i Y_i\end{aligned}$$

Where,

$$c_i = \frac{x_i}{\sum_{j=1}^n x_j^2}$$

Also, note that

$$Var[Y_i] = Var[\beta x_i + \epsilon_i] = Var[\epsilon_i] = \sigma^2$$

Then, by independence of the Y_i 's,

$$\begin{aligned} Var[\hat{\beta}] &= Var \left[\sum_{i=1}^n c_i Y_i \right] \\ &= \sum_{i=1}^n c_i^2 Var[Y_i] \\ &= \sum_{i=1}^n \left(\frac{x_i}{\sum_{j=1}^n x_j^2} \right)^2 \sigma^2 \\ &= \sigma^2 \frac{\sum_{i=1}^n x_i^2}{\left(\sum_{j=1}^n x_j^2 \right)^2} \\ &= \frac{\sigma^2}{\sum_{j=1}^n x_j^2} \end{aligned}$$

Thus, then the standard error of $\hat{\beta}$ is given by

$$\sigma_{\hat{\beta}} = \sqrt{\frac{\sigma^2}{\sum_{j=1}^n x_j^2}}$$

(iii)

Since $\hat{\beta}$ is a linear combination of (i) independent, and (ii) normal random variables,

$$\hat{\beta} = \sum_{i=1}^n c_i Y_i$$

$\hat{\beta}$ is also normally distributed,

$$\hat{\beta} \sim N\left(\beta, \frac{\sigma^2}{\sum_{j=1}^n x_j^2}\right)$$